Temporal Structure, Signed Opinion Dynamics, and Targeted Interventions

Experimental Report and Team Handoff

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19 October 2026; corrected 21 October 2026

Abstract

This report describes the temporal-network contribution to the team's Reddit project. We analyze the timing and sentiment of inter-subreddit hyperlinks, compare signed Non-Consensus Opinion (NCO) on static and evolving graphs, and evaluate timing controls and stubborn-node interventions (ICO). The resolution and ICO results were reverified on 21 October 2026; empirical diagnostics remain from 18 October. Reddit interactions are observed. Binary opinions and intervention outcomes are simulated, not measured beliefs or misinformation.

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1 Research questions and scope

Our work asks: (i) what temporal patterns exist in inter-subreddit interactions? (ii) does representing the graph as evolving weekly snapshots change signed NCO outcomes relative to static aggregation? (iii) does the *actual allocation of event timestamps* matter beyond the lifetime topology? and (iv) does interaction timing affect the efficacy of targeted stubborn-node interventions? The experimental suite includes network characterization, data-driven modeling, matched null models, and paired uncertainty estimates.

2 Dataset, graph representation, and conventions

2.1 Observed input

We use the SNAP Reddit Hyperlink Network, specifically the body hyperlink event log. The complete log contains 286,561 directed timestamped interactions. Restricting to the team's selected 10-core yields **2,896 subreddits**, **159,180 directed events**, and **58,466 lifetime aggregated undirected edges**. All experiments retain the same `nodes.json` ordering and full 2,896-node set, including temporarily inactive nodes. A directed event identifies a source subreddit, a target subreddit, a timestamp, and a sentiment sign. The label $+1$ means *non-negative* (including neutral); -1 means negative. These labels describe hyperlinks, not ground-truth opinions or misinformation.

2.2 Signed temporal snapshots and simulator

Events are grouped into non-overlapping weekly windows (also tested biweekly and monthly). Within each window, directions are merged and repeated interactions between a pair are collapsed to the majority sign; equal positive and negative counts give signed weight zero while the edge remains in the unsigned topology. Let $W^{(t)}$ be the signed matrix in window t, and $\sigma_i^{(t)} \in \{-1, +1\}$ the *simulated* node opinion. The synchronous update is

$$h_i^{(t)} = \sigma_i^{(t)} + \sum_j W_{ji}^{(t)} \sigma_j^{(t)}, \qquad \sigma_i^{(t+1)} = \begin{cases} +1 & h_i^{(t)} > 0 \\ -1 & h_i^{(t)} < 0 \\ \sigma_i^{(t)} & h_i^{(t)} = 0. \end{cases} \tag{1}$$

One update occurs per window, over the complete calendar sequence. The initial fraction f denotes nodes initialized to -1, usually $f = 0.1, 0.2, \ldots, 0.9$. Initial configurations and random seeds are paired across model/control comparisons. The fixed static baseline reuses a single lifetime-aggregated signed matrix at each algorithmic iteration; its iterations are *not* calendar weeks.

2.3 Metrics—important differences

Let $p_\pm$ be the final fractions of nodes holding ± 1. The *minority fraction* is $q = \min(p_+, p_-)$, and absolute magnetization is $|m| = |p_+ - p_-| = 1 - 2q$. These two metrics are mathematically redundant. The *minority LCC fraction* is the size of the largest minority-opinion connected component divided by the *number of minority nodes*; it measures how concentrated the surviving minority is in one connected cluster. The team's ICO metric instead uses $g_- = S_{-\mathrm{GCC}}/N$: the size of the largest -1 component divided by *all* 2,896 nodes. These measures must not be interchanged. Component calculations use the fixed lifetime unsigned core. Total switching counts opinion changes over algorithmic updates. Φ is the intervention fraction of nodes pinned to $+1$; success is $g_- < 0.5$, and Φ_c is the first successful *tested* budget, with zero when the baseline already meets this criterion.

3 Controlled randomizations and what they preserve

3.1 Global and within-source timestamp nulls

Global shuffle: permute the timestamp column across all original events, leaving each event's directed source, target, and sentiment unchanged. This exactly preserves the complete lifetime multigraph, all node lifetime degrees and event counts, all lifetime signed aggregate edges, the timestamp multiset, and therefore calendar-wide spikes and event counts per week. It does *not* preserve node-specific or edge-specific weekly activity, weekly degrees, dyad timing, or incident-edge temporal correlations. Thus it does *not* remove every kind of burstiness: global platform-wide spikes remain.

Within-source shuffle: permute event timestamps among events with the same source subreddit. This additionally preserves each source's time-resolved outgoing activity, while disturbing when that source contacts particular targets. Directed event identities and sentiment remain unchanged. Both nulls use the same simulation rule as the real temporal graph.

Sentiment nulls: separate analyses redistribute sign labels globally, within source, or within undirected dyad. The within-dyad null retains each pair's overall positive/negative event composition and is the strictest control used for sign persistence. A timestamp shuffle and a sign shuffle address different hypotheses.

3.2 Preservation audit

The audit confirms preservation of lifetime degrees, signed edges, and global event counts per window on the 2,896-node, 58,466-edge core. Corrected weekly, biweekly, and monthly snapshots contain 174, 87, and 41 windows; each retains all 159,180 events. The old biweekly bins omitted 79,644 events. Weekly node and edge activity need not be preserved. Permutations test disrupted properties, not a unique causal mechanism.

4 Empirical temporal network characterization

4.1 Source-level activity concentration and burstiness

Metric	Real	Global timestamp null
Mean active weeks per source	29.303	34.717
Inter-event-time burstiness	0.05390	-0.05283
Weekly Fano factor (variance/mean)	2.3308	1.0060

For each of these metrics, 100 null shuffles yielded an empirical reported two-sided $p = 0.019802$. The real activity series are concentrated in fewer weeks and have substantially larger count variance than the global null.

Interpretation. Subreddit interaction activity is unevenly concentrated in time, beyond what is explained merely by the global calendar histogram. This is empirical network characterization, not an outcome of simulated opinion dynamics. The null preserves global activity spikes, so this is principally a statement about *source-level* burstiness.

4.2 Correlated activity on incident edges

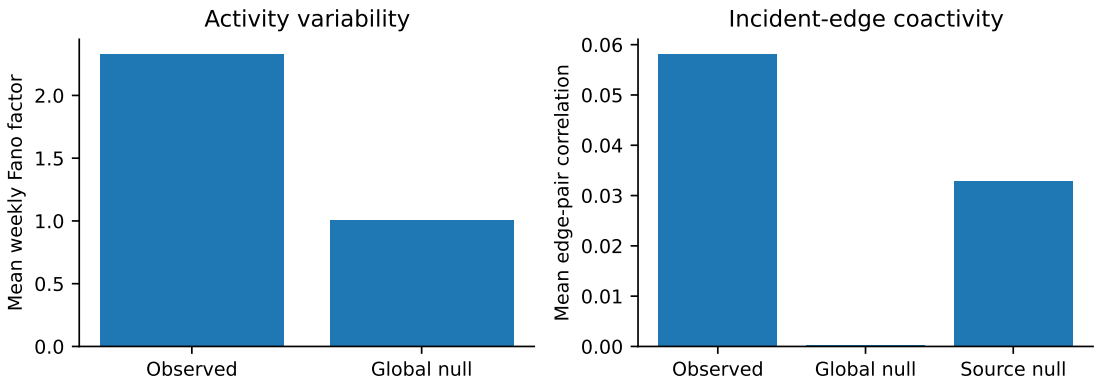


Figure 1: Activity dispersion and incident-edge correlations (observed vs. temporal nulls).

For sampled pairs of connections sharing a subreddit, mean Pearson correlation of weekly interaction counts was 0.058165 in the real data, 0.000179 under global shuffling, and 0.032825 under within-source shuffling. The dedicated 100-shuffle comparisons both reported empirical two-sided $p = 0.019802$.

Interpretation. Adjacent connections are weakly but detectably co-active in real time. Preserving the source's activity schedule accounts for part of this correlation but does not reproduce all of it. The effect is small in absolute magnitude. In a majority-update model, simultaneous neighbor activity is a plausible mechanism for changed influence, but our experiments do not establish this correlation as the cause of NCO differences.

4.3 Directed seven-day reciprocity by sentiment

A directed event $A \to B$ is reciprocated if a later $B \to A$ event of the same sentiment class occurs strictly within seven days. We calculate the rate among eligible initiating events and compare with matched timing nulls.

Sentiment	Real rate	Global null	Source null	Real/global
Negative	0.3888%	0.2045%	0.2209%	1.90×
Non-negative	5.3369%	2.7535%	3.3630%	1.94×

Each real-versus-null comparison had reported Monte Carlo two-sided $p = 0.019802$ with 100 permutations. The real/source enrichment is 1.76× for negative and 1.59× for non-negative references.

Interpretation. Directed reciprocal references are concentrated in time for *both* sign classes. Negative reciprocity is not uniquely enriched under the global null. Negative events have a much lower absolute reciprocity rate. These associations do not establish retaliation, coordination, or causal responses. The NCO simulator symmetrizes edges, so directed reciprocity is a distinct empirical characterization result, not a demonstrated NCO mechanism.

4.4 Sign persistence and increasingly constrained controls

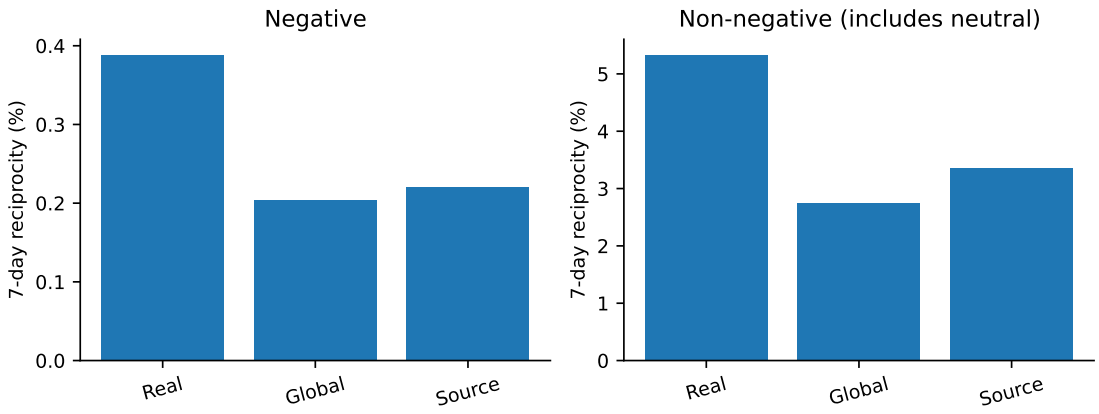


Figure 2: Observed directed seven-day reciprocity compared with timestamp-based nulls; note the different vertical scales for the two sentiment groups.

Define a dyad's weekly sign as positive, negative, or mixed, and consider transitions across consecutive active observations. Observed $+ \to +$ persistence is 0.912888; observed $- \to -$ persistence is 0.193548. Under within-dyad sentiment randomization the corresponding null means are 0.908920 and 0.183037. The positive-to-positive difference is +0.003967, empirical $p = 0.029851$, whereas negative-to-negative excess is +0.010511, $p = 0.298507$. Mixed-to-mixed is observed 0.067327 versus null 0.068282, $p = 0.955224$.

Interpretation. Apparent excess negative-sign persistence under simpler sign controls largely disappears when each dyad's overall sentiment mix is preserved. The data do not provide convincing evidence for an additional negative temporal-persistence effect beyond pair-level heterogeneity under this strict null. The positive excess is very small and comparisons are uncorrected for multiple testing.

5 Temporal signed NCO outcomes

5.1 Experiment A: static lifetime graph versus real weekly network

Both models use identical initial opinions and the same signed self-vote update. Static NCO uses the full lifetime aggregate matrix at every iteration, whereas temporal NCO uses the current week's matrix once per week. The fixed-update-budget comparison runs the static process for as many steps as temporal windows. An independent static stopping audit found that *all 180 static runs* (20 per initial fraction) reached a fixed point, with mean convergence steps $3.00, 3.20, 3.65, 4.75, 8.20, 4.25, 3.75, 3.15, 3.00$ for $f = 0.1, \ldots, 0.9$, respectively. No cycles or update-limit outcomes occurred in that audit.

Initial f	Static final minority	Real temporal minority	Difference (pp)
0.1	0.36%	3.45%	+3.09
0.2	0.40%	4.58%	+4.18
0.3	0.45%	5.70%	+5.25
0.4	0.88%	7.34%	+6.46
0.5	3.85%	10.51%	+6.66
0.6	0.76%	7.14%	+6.38
0.7	0.55%	5.71%	+5.17
0.8	0.41%	4.60%	+4.19
0.9	0.37%	3.52%	+3.15

Paired bootstrap 95% intervals for the temporal-minus-static minority difference exclude zero for all nine fractions in the reported 20-initialization experiment.

Interpretation. Temporal signed NCO retains more minority opinions at every tested initialization, most visibly near balanced $f = 0.5$. However, this comparison mixes timing, per-update edge availability, and repeated exposure to a much denser lifetime graph. It cannot by itself establish an ordering-only effect. The separate convergence audit supports treating the static terminal opinions as fixed points for its audited configurations.

5.2 Experiment B: real weekly timing versus globally shuffled timing

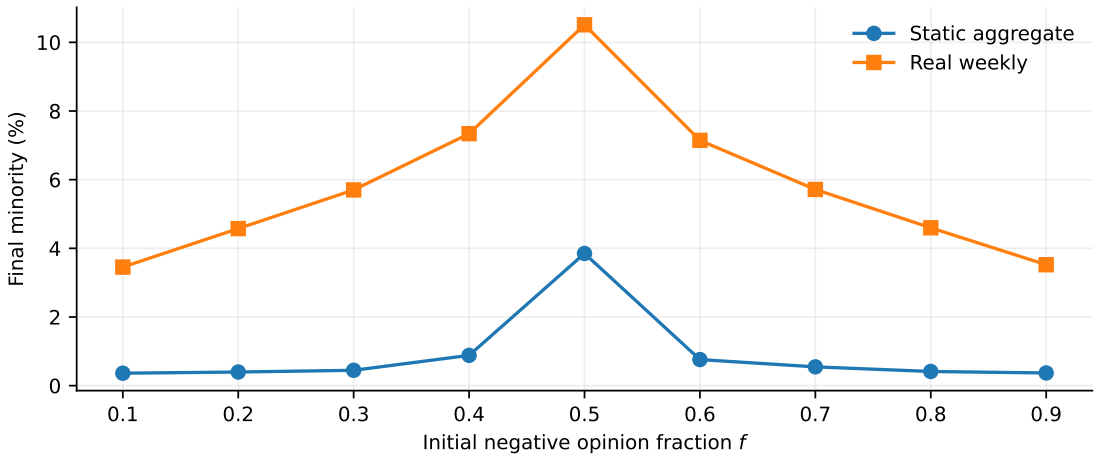


Figure 3: Static versus real temporal NCO: final minority proportion across all initial fractions.

Both conditions use weekly snapshots and the same original event endpoints, signs, and aggregate graph. Only event timestamps are reassigned in the null. The final minority values are:

Initial f	Real minority	Shuffled minority	Real − shuffled (pp)
0.1	3.45%	3.06%	+0.40
0.2	4.58%	5.29%	-0.72
0.3	5.70%	7.61%	-1.91
0.4	7.34%	10.87%	-3.53
0.5	10.51%	16.69%	-6.18
0.6	7.14%	10.61%	-3.46
0.7	5.71%	7.67%	-1.96
0.8	4.60%	5.30%	-0.70
0.9	3.52%	3.05%	+0.47

At $f = 0.5$, real absolute magnetization is 0.7898 and shuffled magnetization 0.6663 (equivalently 10.51% versus 16.69% minority). Differences in magnetization are strongly initialization-dependent; the sign reverses at very imbalanced f values.

Interpretation. Actual timing assignments matter for model predictions even when the complete lifetime graph is held fixed. Near balanced starts, observed timing yields stronger majority dominance than global timestamp shuffling. The null disrupts multiple temporal properties simultaneously; no unique explanation such as bursts or co-activity has been causally isolated.

5.3 Switching, node-level disagreement, and time-resolution sensitivity

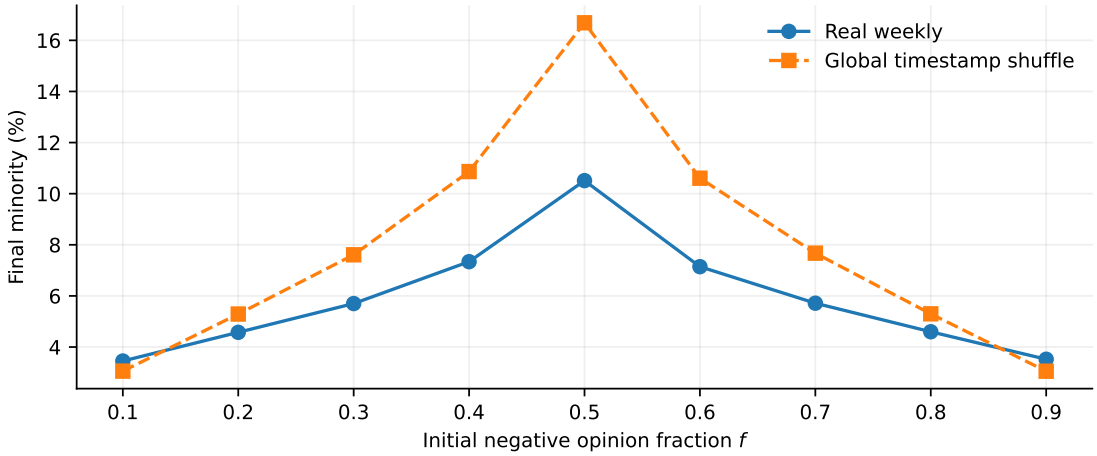


Figure 4: Real versus globally timestamp-shuffled temporal NCO: final minority proportion.

At $f = 0.5$, total simulated changes were approximately 3,016 (static), 3,632 (real temporal), and 4,031 (globally shuffled temporal). Real-versus-shuffled switching changes sign with initialization: at $f = 0.1$ real has ~ 387 *more* switches, while at $f = 0.5$ real has ~ 399 *fewer*. Thus timing does not universally increase or decrease volatility. Static iterations are not calendar weeks, and different snapshot lengths give different numbers of update opportunities.

Final individual-node opinion assignments differed between static and temporal outcomes for roughly 3.5–7.6% of nodes at most initial fractions, rising to about 22.9% at $f = 0.5$ (approximately 662 nodes on average; wide bootstrap interval around 12.0–34.7%). This descriptive measure partly reflects differences in minority size or even which label becomes the majority, so it is not a standalone influence mechanism.

At f = 0.5, real-minus-shuffled switching is -367 (95% CI [-470, -266]) weekly, -307 ([-395, -216]) biweekly, and -345 ([-450, -239]) monthly. Corrected biweekly bins retain all 159,180 events. Intervals resample 20 initializations and 20 shared timelines (2,000 draws). Raw counts differ with update opportunities; final +1-share intervals include zero.

5.4 Experiment C: structural cohesion of the final minority

Minority LCC fraction, on the fixed lifetime unsigned network, is normalized by the number of *minority nodes*. In an independent temporal fragmentation experiment:

Initial f	Real minority LCC	Global shuffle LCC	Within-source LCC
0.3	34.3%	18.5%	22.1%
0.5	47.1%	47.6%	43.0%
0.7	33.9%	18.7%	22.1%

The real temporal minority is more concentrated in its largest cluster at $f = 0.3$ and $f = 0.7$, with reported paired-bootstrap intervals excluding zero for the real-versus-global differences; the effect is absent or reversed around $f = 0.5$. Real timing also produced fewer raw minority components than the global shuffle in the corresponding comparison, although component counts depend on how many minority nodes survive.

Interpretation. Interaction timing influences how surviving minority opinions are positioned on the network, in addition to how many survive. The ratio is sensitive to the total size of the minority, so do not treat a larger ratio as automatically indicating a larger absolute minority cluster. This statistic is *not* the team's g_- ICO metric.

6 Temporal ICO: targeted stubborn-node interventions

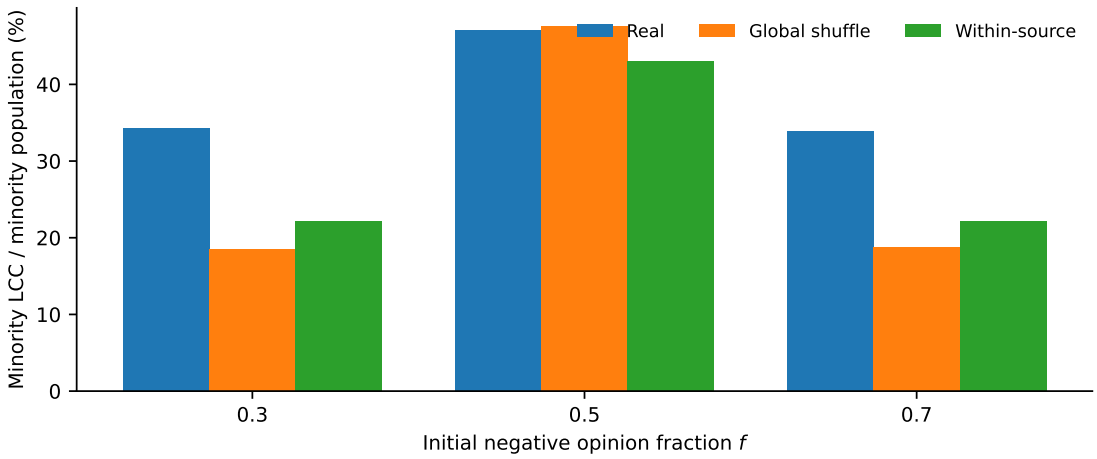


Figure 5: Relative size of the largest surviving minority cluster on the fixed lifetime graph.

6.1 Design and objectives

A chosen set of nodes is pinned to opinion +1 throughout temporal signed updates. Budget Φ denotes their fraction of the 2,896 nodes. The runner currently evaluates *random* and *lifetime degree* targeting; signed-leverage targeting is implemented as an input option but awaits Georgi's ranking export. The intervention is introduced at time zero, consistent with the existing team ICO convention; a direct static-versus-temporal ICO outcome comparison still requires ensuring matched metrics and stopping conventions. We use g_-, not minority fraction, as the intervention objective. The key f values are $0.3, 0.5, 0.7$, across budget grid $\Phi \in \{0, .01, .03, .05, .10, .15, .20, .25\}$. Full runs pair 20 opinion initializations with 20 shared timestamp shuffles. The 2,000 bootstrap draws resample both initializations and timeline IDs, preserving matched comparisons.

6.2 Experiment D: intervention budget curves and strategy effect

At $f = 0.7$, the initial negative opinion dominates and the no-intervention opposing GCC is $g_- = 0.9402$ for real timing. Mean real-temporal outcomes are:

Budget Φ	Degree targeting g_-	Random targeting g_-
0	0.9402	0.9402
.01	0.2384	0.9262
.03	0.0949	0.8956
.05	0.0638	0.8643
.10	0.0234	0.7414
.15	0.0114	0.3790
.20	0.0100	0.0343
.25	0.0078	0.0203

At $\Phi = .05$, the paired real degree-minus-random difference is -0.80055, 95% bootstrap CI [-0.80931, -0.79090].

Interpretation. Degree-targeted stubborn nodes dramatically reduce the simulated -1 GCC relative to randomly targeted nodes. This demonstrates a strong relationship between lifetime topology and intervention leverage in this particular signed temporal model. It is not a measured effect of real fact-checking and does not identify real misinformation.

6.3 Experiment E: real versus shuffled timing for intervention efficacy

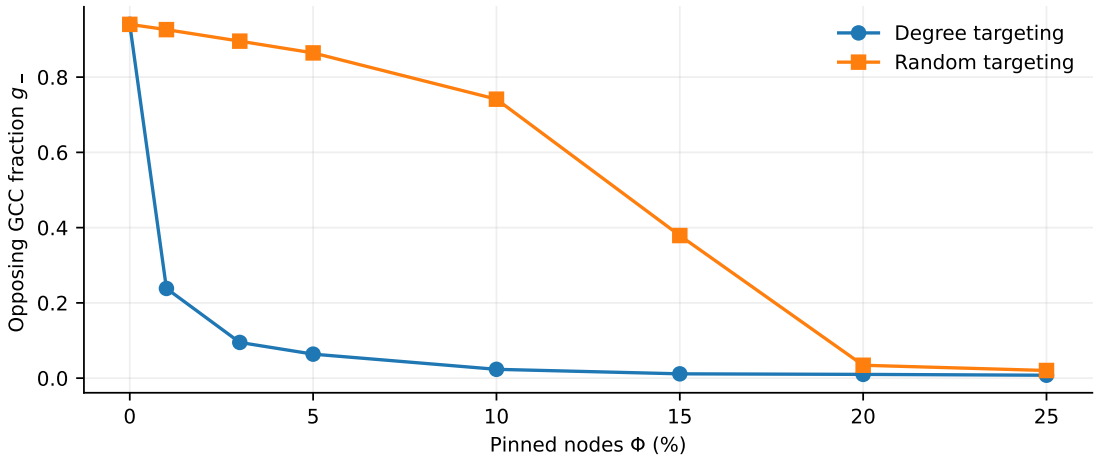


Figure 6: Temporal ICO: full real-timing budget response at $f = 0.7$.

At $f = .7, \Phi = .05$:

Strategy	Real g_-	Shuffled g_-	Real − shuffled	95% paired CI
Degree	0.06379	0.14246	-0.07867	[-0.08572,-0.07117]
Random	0.86435	0.84619	+0.01816	[+0.01492,+0.02197]

Compared with shuffled timing, real timing leaves a smaller negative GCC under degree targeting and a slightly larger one under random targeting. Paired bootstrap draws resample the 20 initializations and 20 shared timelines; the 400 null evaluations are not independent runs.

Interpretation. Temporal organization changes the relative performance of interventions even with identical lifetime topology and targeting scores. The interaction timing may change when targeted nodes can influence their neighbors, but the specific causal mechanism was not isolated.

6.4 Experiment F: per-run success and critical budget

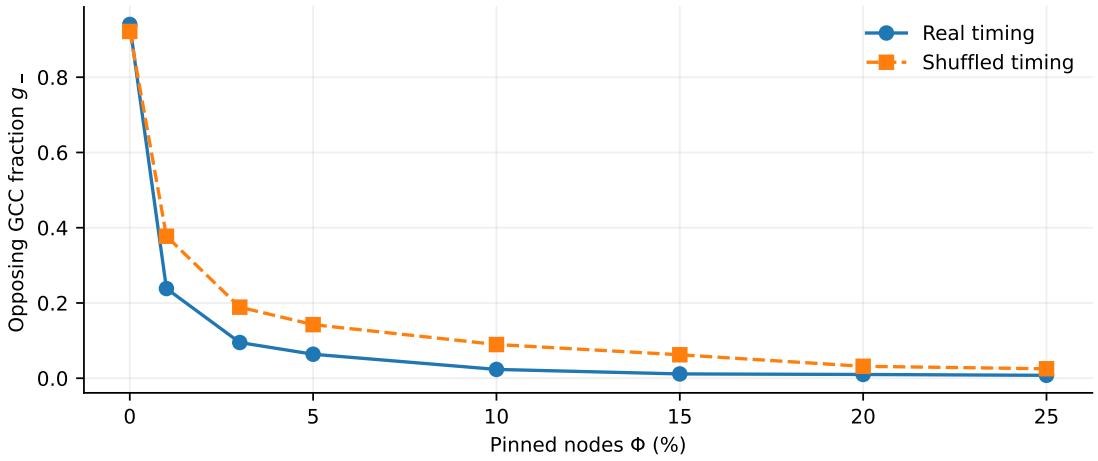


Figure 7: Temporal ICO: degree targeting on real versus shuffled weekly networks.

Success is $g_- < .5$. At $f = .7$, real degree targeting succeeds in **20/20 runs at** $\Phi = .01$, the smallest nonzero tested budget, while random targeting succeeds in 0/20 at $\Phi = .01$, 1/20 at .10,

10/20 at .15, and 20/20 at .20. Real random Φ_c is .10 in one run, .15 in ten, and .20 in nine (median .15); degree $\Phi_c = .01$ in all 20. At $f = .5$, 55% of real no-intervention runs already satisfy the threshold, reflecting variability in which label wins. At $f = .3$, the negative GCC is below .5 without intervention, so $\Phi_c = 0$ and this condition is not informative about crossing the chosen threshold.

Shuffled $f = .7$ degree targeting also crosses the .5 threshold at $\Phi = .01$ across sampled runs, although its residual g_- remains larger than for real timing. At $\Phi = .15$, real random targeting succeeds in 50% of runs and shuffled random targeting in about 70.25% of paired evaluations.

Interpretation. A binary success threshold can hide substantial differences in continuous g_-. A critical budget on a coarse grid is not an exact phase-transition point, and these success rates are conditional on the tested initializations and randomization ensemble. The 400 shuffled evaluations are not 400 independent opinion initializations.

7 Reproducibility, cautions, and method validity

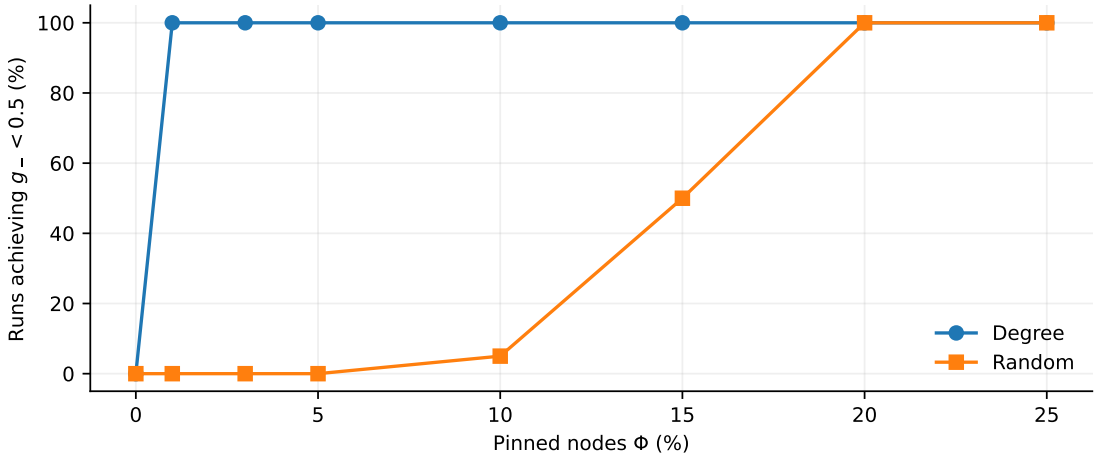


Figure 8: Temporal ICO: fraction of runs meeting the intervention threshold $g_- < 0.5$.

- **Empirical versus simulated:** Reddit timestamps, sources, destinations, and event sentiment are observations. Node binary opinion, consensus, switching, and pinned +1 intervention effects are simulated. The negative event label does not mean misinformation; +1 event sentiment is not equivalent to the modeled +1 opinion.
- **Network reduction:** Results concern the chosen 10-core; core extraction, temporal symmetrization, majority-sign collapse, and one-update-per-window policy are modeling decisions. Zero-sign ties remain unsigned edges with no signed influence.
- **Static versus temporal:** Both are valid representations but differ in edge exposure. Equal numbers of updates do not imply equal contact opportunities. Our separate audit found static steady-state convergence in every tested run.
- **Null validity:** Full-lifetime signed topology and degree preservation are verified. Global timestamp shuffling does not destroy calendar-wide global event spikes and changes multiple node/edge-level temporal properties at once. No single null comparison proves a unique mechanism.
- **Statistical uncertainty:** ICO and resolution intervals resample 20 initializations and 20 shared timelines with 2,000 paired bootstrap draws. Diagnostic p-values use finite permutations (e.g., 0.019802 with 100 shuffles); multiple testing was not family-wise or FDR adjusted.

- **Mechanistic claims:** Burstiness, incident-edge coactivity, and reciprocity are plausible explanatory properties but have not been isolated as causal mediators of the NCO/ICO differences.

8 What remains, who it depends on, and handoff

Item	Depends on	Action / status
Temporal signed-leverage ICO	**Georgi**	Obtain node-aligned signed-leverage ranking CSV (`node,signed_leverage`), then rerun temporal ICO on the agreed budget grid. The runner already accepts this ranking; it does not invent scores.

9 Recommended messages for the final presentation

1. **Data first:** Reddit activity is burstier at the community level and incident connections are more co-active than timestamp nulls predict.

2. **Representation matters:** Static aggregation gives stronger final consensus than real weekly signed NCO, but this includes different edge-exposure assumptions.

3. **Timing itself matters:** Holding all lifetime interactions and aggregated topology fixed, global timestamp shuffling changes both opinion outcomes and targeting effectiveness; the direction can depend on initial f and intervention strategy.

4. **Network topology and intervention:** Degree targeting decisively reduces the simulated -1 GCC relative to random targeting at $f = .7$. Real timing makes this degree intervention more effective than its timestamp-shuffled counterpart under the tested metric.

5. **Appropriate controls matter:** Directed temporal reciprocity is enriched for both event sentiment classes, while apparent negative persistence loses support under a dyad-conditioned sign null.

Bottom line: An aggregate network is insufficient to reproduce all predictions of this signed temporal opinion and intervention model. The actual allocation of timestamped interactions affects the outcomes, but these are results of a specified simulation on a real interaction network, not causal evidence about real misinformation or belief change.

Input/result provenance

Original characterization, sign-persistence, reciprocity, and robustness results come from the team's 17-18 October 2026 runs. On 21 October, fixes at commit d81bc6e were verified against the same 2,896-node core and 159,180 events. Reruns covered 180 paired static/weekly initializations, the complete 20,160-run ICO grid, and resolution comparisons with 20 and 100 timestamp shuffles. Corrected intervals reported here use 20 initializations, 20 shared timelines, and 2,000 bootstrap draws; all 840 earlier ICO spot-checks matched. The empirical diagnostic permutation suites were not rerun. The corrections preserve the main conclusions and do not establish causal effects on real beliefs.